EAB 770 - Chapter 3

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1 Chapter 3: Sets of 2×2 tables

1.1 Outline

- Cochran-Mantel-Haenszel test
- Measures of Association
  - Common Odds ratio
  - Breslow-Day test
  - Zelen's Test

2 Cochran-Mantel-Haenszel Statistic

2.1 Sets of 2×2 tables

- Consider a multi-location RCT
  - Different locations will have different contingency tables
  - We want to consolidate **all** data and account for the location effect in testing for the association between the treatment and outcome.

Table 3.1 Respiratory Improvement

Center	Treatment	Yes	No	Total
1	Test	29	16	45
1	Placebo	14	31	45
Total		43	47	90
2	Test	37	8	45
2	Placebo	24	21	45
Total		61	29	90

2.2 Stratified Analysis

Stratified analysis examines the association between two variables while adjusting for the effect of other covariates.

> **Note**
>
> The variable that define the strata may be an explanatory variable, location, or classification variable. Each contingency table corresponds to one level of the explanatory variables (stratum). The contingency tables at each level of the stratification variable are called **partial tables**.

2.3 Conditional Associations

Associations in partial tables are called conditional associations because it is conditional on a particular level of the stratification variable.

> **Note**
>
> We can use the methods from the previous chapters to test for conditional associations at each level of the stratification variable. Odds ratios can be calculated for each partial table.

2.4 Marginal Associations

Marginal associations are associations in marginal tables.

Marginal Table

A marginal table contains the summed frequencies of the partial tables across the stratification variable. Suppose the frequency in the cell (i, j) in the kth partial table is n_{ijk}. The marginal table will have the following value for the cell (i, j):

$$n_{ij} = \sum_{k=1}^{K} n_{ijk}$$

Tip

Conditional associations examine two-way associations at each strata, but marginal associations examine the overall two-way associations by averaging over all strata. Interpreting marginal association in the presence of differing conditional associations is **NOT** recommended.

2.5 Independence

Marginal Independence

Marginal independence implies that the there is no association in the marginal table. When there is marginal independence, the marginal odds ratio (OR_{XY} or θ_{XY}) is equal to 1. In general, for any $s \times r$ table, marginal independence implies that all odds ratios that can be formed using any two levels of the variables will be equal to 1.

Conditional Independence

Conditional independence implies that there is no association between the variables X and Y in any of the partial tables that are conditional on the stratification variable levels. Conditional independence implies that the conditional odds ratios ($OR_{XY|Z}$ or $\theta_{XY|Z}$) in all the partial tables are equal to 1.
Conditional dependence implies that there is an association in at least one of the partial tables. Conditional independence does not imply marginal independnece and vice versa.

Tip

In the symbolic convention, X and Y are the variables being tested for association, Z is the stratification variable/covariate.

2.6 Homogeneous Association

When the conditional associations between X and Y are the same across all partial tables in each strata of Z, we have **homogeneous association** between X and Y. For K strata,

$$\theta_{XY|Z_1} = \theta_{XY|Z_2} = ... = \theta_{XY|Z_K}$$

By definition, the following equalities should also hold:

$$\theta_{XZ|Y_1} = \theta_{XZ|Y_2} = ... = \theta_{XZ|Y_K}$$

$$\theta_{YZ|X_1} = \theta_{YZ|X_2} = ... = \theta_{YZ|X_K}$$

2.7 Notes

> **Simpson's Paradox**
>
> Simpson's paradox corresponds to the scenario where the marginal association is completely different from the conditional association.

Although conditional independence implies homogeneous association (because $\theta_{XY|Z} = 1$ for all strata), homogeneous association does not necessarily imply conditional independence. For homogeneous association to hold, the odds ratios have to be equal but are not required to be equal to 1.

3 Inferential Statistics

3.1 Mantel-Haenszel Test

Recall the Mantel-Haenszel Chi-Square statistic

$$Q = (n_{11} - m_{11})^2 / v_{11}$$

where m_{11} is the expected value of the n_{11} and v_{11} is the variance of n_{11} based on the hypergeometric distribution assumption.

> **Note**
>
> If there are q 2×2 tables, each table will have a value for Q.

3.2 Cochran-Mantel-Haenszel Test

The Cochran-Mantel-Haenszel test assesses the presence of conditional association of X and Y after adjusting for the strata Z. If n_{j11} is the n_{11} value of the jth contingency table, the Cochran-Mantel-Haenszel statistic, Q_{MH} can be written as:

$$Q_{MH} = \frac{\left[\sum_{j=1}^{q} n_{j11} - \sum_{j=1}^{q} m_{j11}\right]^2}{\sum_{j=1}^{q} v_{j11}}$$

> **Note**
>
> Q_{MH} is often called an **average partial association statistic**.

3.3 Cochran-Mantel-Haenszel Test: Hypotheses

The Cochran-Mantel-Haenszel test tests the following hypotheses:

- H_0: Given the stratification variable, there is conditional independence such that all conditional odds ratios are equal to 1.
- H_a: At least one conditional odds ratio is not equal to 1.

3.4 Mantel-Haenszel Statistic

The statistic Q_{MH} relies on detecting patterns of the differences of proportions in each table.

> **Note**
>
> The statistic Q_{MH} approximately follows a chi-square distribution with one degree of freedom if the **combined** random sample sizes are large. This means that individual cell counts and isolated table sample sizes may be small.
> The **Mantel-Fleiss criterion** can be used to check if the chi-square assumption holds. The Mantel-Fleiss criterion should be greater than 5.

> **Warning**
>
> There is no function in `R` that can calculate the Mantel-Fleiss criterion, but `SAS` will provide the estimate of the Mantel-Fleiss criterion. The Mantel-Fliess criterion can be calculated using the following formula for a total of K strata:

$$MF = min\left\{\sum_{h=1}^{K}[m_{h11} - max(0, n_{h1+} - n_{h+2})], \sum_{h=1}^{K}[min(n_{h+1}, n_{h1+}) - m_{h11}]\right\}$$

3.5 Cochran-Mantel-Haenszel Test: SAS

- The `cmh` option in the `tables` statement produces the Mantel-Haenszel statistic values. There will be three Mantel-Haenszel values included in the output.
- Nonzero Correlation
- Row Mean Scores Differ
- General Association

> **Note**
>
> For 2×2 tables, the scores and degrees of freedom do not differ for these three MH-statistics. These measures will be distinguished for larger contingency tables.

> **Note**
>
> The `mf` option in the `tables` statement produces the Mantel-Fleiss criterion. Recall that the Mantel-Fleiss criterion (MF) should be greater than 5 for the chi-square distributional assumption to be valid.

3.6 Cochran-Mantel-Haenszel Test: R

Just like the previous chapter, we can use the `mantelhaen.test()` function in `R`.

```
contingency_table <- xtabs(count~row_var+column_var+strata_var, data=dat)
# OR

contingency_table <- xtabs(~ row_var+column_var+strata_var, data=dat) # for non-aggregated da


mantelhaen.test(contingency_table, correct = FALSE) #no correction gives the same result as S
```

To explore conditional associations, we can use the `CMHtest()` function in the `vcdExtra` package in `R`.

```
CMHtest(contingency_table,
        strata=strata_var,
        overall=T)
```

3.7 Example

The table shows the results of a study about how general daily stress affects one's opinion on a proposed new health policy. Participants were sampled from urban and rural areas.

3.7.1 Question

- Is there enough evidence that there is an association between the opinion and stress levels of residents after controlling for residential area? Use =0.05 as the significance level.

Table 3.3 Health Policy Opinion Data

Residence	Stress	Favorable	Unfavorable	Total
Urban	Low	48	12	60
Urban	High	96	94	190
	Total	144	106	250
Rural	Low	55	135	190
Rural	High	7	53	60
	Total	62	188	250

3.7.2 SAS Code

```
1  data stress;
2  input region $ stress $ outcome $ count @@;
3  datalines;
4  urban low f 48 urban low u 12
5  urban high f 96 urban high u 94
6  rural low f 55 rural low u 135
7  rural high f 7 rural high u 53
8  ;
9  proc freq order=data data=stress;
10 weight count;
11 tables region*stress*outcome / all chisq cmh (mf) nocol nopct;   ①
12 run;
```

① `CMH` includes the Mantel-Haenszel statistics. `MF` gives the Mantel-Fleiss condition;

3.7.3 R Code

```
library(vcdExtra)
```

```
Warning: package 'vcdExtra' was built under R version 4.6.1
```

```
Loading required package: vcd
```

```
Warning: package 'vcd' was built under R version 4.6.1
```

```
Loading required package: grid
```

```
Loading required package: gnm
```

```
Warning: package 'gnm' was built under R version 4.6.1
```

```
Registered S3 methods overwritten by 'vcdExtra':
  method      from
  pairs.table vcd
  print.Kappa vcd
```

```
Attaching package: 'vcdExtra'
```

```
The following objects are masked from 'package:vcd':

    pairs_diagonal_mosaic, woolf_test
```

```
dat  <- expand.grid(Outcome = c("Favorable","Unfavorable"),
                    Residence=c("Urban","Rural"),
                    Stress = c("Low","High")
                   )
dat$count <- c(48,12,55,135,96,94,7,53)

dat_cc <- xtabs(count ~  Stress + Outcome + Residence  ,data=dat)
dat_cc
```

```
, , Residence = Urban

      Outcome
Stress Favorable Unfavorable
  Low         48          12
  High        96          94

, , Residence = Rural

      Outcome
Stress Favorable Unfavorable
  Low         55         135
  High         7          53
```

```r
mantelhaen.test(dat_cc,correct=F)
```

```
    Mantel-Haenszel chi-squared test without continuity correction

data:  dat_cc
Mantel-Haenszel X-squared = 23.05, df = 1, p-value = 1.578e-06
alternative hypothesis: true common odds ratio is not equal to 1
95 percent confidence interval:
 2.066744 6.069367
sample estimates:
common odds ratio
         3.541726
```

```r
CMHtest(dat_cc,
        strata="Residence",
        overall=T)
```

```
$`Residence:Urban`
Cochran-Mantel-Haenszel Statistics for Stress by Outcome
    in stratum Residence:Urban

                AltHypothesis  Chisq Df       Prob
cor        Nonzero correlation 16.155  1 5.8367e-05
rmeans  Row mean scores differ 16.155  1 5.8367e-05
cmeans  Col mean scores differ 16.155  1 5.8367e-05
general    General association 16.155  1 5.8367e-05
```

```
$`Residence:Rural`
Cochran-Mantel-Haenszel Statistics for Stress by Outcome
    in stratum Residence:Rural

                AltHypothesis  Chisq Df      Prob
cor        Nonzero correlation 7.2724  1 0.0070022
rmeans  Row mean scores differ 7.2724  1 0.0070022
cmeans  Col mean scores differ 7.2724  1 0.0070022
general    General association 7.2724  1 0.0070022


$ALL
Cochran-Mantel-Haenszel Statistics for Stress by Outcome
    Overall tests, controlling for all strata

                AltHypothesis Chisq Df       Prob
cor        Nonzero correlation 23.05  1 1.5783e-06
rmeans  Row mean scores differ 23.05  1 1.5783e-06
cmeans  Col mean scores differ 23.05  1 1.5783e-06
general    General association 23.05  1 1.5783e-06
```

Calculating the Mantel-Fleiss manually. The function below uses the `mantel_fleiss(contingencytable,k)` function where `contingencytable` corresponds to the contingency table in the problem and `k` is the number of strata.

```r
mantel_fleiss <- function(cc,k){
  sum_expected =0
  sum_max = 0
  sum_min =0
  for (h in 1:k){
    nh1p = rowSums(cc[,,h])[1]
    nhp2 = colSums(cc[,,h])[2]
    nhp1 = colSums(cc[,,h])[1]
    m11 = nh1p*nhp1/sum(cc[,,h])
    sum_expected = sum_expected + m11
    sum_max = sum_max + max(0,nh1p-nhp2)
    sum_min = sum_min + min(nhp1,nh1p)

  }
  min(sum_expected-sum_max,sum_min-sum_expected)
```

```
}

mantel_fleiss(dat_cc,k=2)
```

```
[1] 40.32
```

3.7.4 Answer

$Q_{MH} = 23.05$, $df = 1$, $p < 0.0001$. There is strong evidence that there is some conditional association between stress level and opinion. Examining each residence level, we see that there is strong evidence of conditional association between stress level and opinion at both rural and urban residences.

3.8 Example

The table shows the results of an RCT for a test treatment designed for respiratory infections. Is there a significant association between the treatment and the outcome after controlling for the center?

- Is the Mantel-Fleiss criterion satisfied?

```
, , Center = 1

         Improvement
Treatment Yes No
  Test     29 16
  Placebo  14 31

, , Center = 2

         Improvement
Treatment Yes No
  Test     37  8
  Placebo  24 21
```

3.9 Exercise

The table shows the results of an RCT for a test treatment designed for respiratory infections. Is there a significant association between the treatment and the outcome after controlling for the center?

3.9.1 Data

```
, , Center = 1

         Improvement
Treatment Yes No
  Test     29 16
  Placebo  14 31

, , Center = 2

         Improvement
Treatment Yes No
  Test     37  8
  Placebo  24 21
```

3.9.2 SAS Code

```
data resp;
length treatment $7.;
input treatment $ improvement $ center $ count;
datalines;
Test Yes 1 29
Placebo Yes 1 14
Test No 1 16
Placebo No 1 31
Test Yes 2 37
Placebo Yes 2 24
Test No 2 8
Placebo No 2 21
;


proc freq data=resp order=data;
weight count;
tables center*treatment*improvement / cmh (MF) nocol norow nopct;
run;
```

3.9.3 R Code

```
dat  <- expand.grid(Improvement = c("Yes","No"),
                    Center=c(1,2),
                    Treatment = c("Test","Placebo")
                   )
dat$count = c(29,16,37,8,14,31,24,21)

dat_cc <- xtabs(count ~  Treatment+Improvement + Center, data=dat)
dat_cc
```

```
, , Center = 1

         Improvement
Treatment Yes No
  Test     29 16
  Placebo  14 31

, , Center = 2

         Improvement
Treatment Yes No
  Test     37  8
  Placebo  24 21
```

```
mantelhaen.test(dat_cc,
                correct=F)
```

```
    Mantel-Haenszel chi-squared test without continuity correction

data:  dat_cc
Mantel-Haenszel X-squared = 18.411, df = 1, p-value = 1.781e-05
alternative hypothesis: true common odds ratio is not equal to 1
95 percent confidence interval:
 2.105716 7.708353
sample estimates:
common odds ratio
         4.028846
```

```r
CMHtest(dat_cc,
        strata="Center",
        types="ALL",
        overall=T)
```

```
$`Center:1`
Cochran-Mantel-Haenszel Statistics for Treatment by Improvement
    in stratum Center:1

                AltHypothesis  Chisq Df      Prob
general    General association 9.9085  1 0.0016452
rmeans  Row mean scores differ 9.9085  1 0.0016452
cmeans  Col mean scores differ 9.9085  1 0.0016452
cor        Nonzero correlation 9.9085  1 0.0016452


$`Center:2`
Cochran-Mantel-Haenszel Statistics for Treatment by Improvement
    in stratum Center:2

                AltHypothesis  Chisq Df      Prob
general    General association 8.5025  1 0.0035465
rmeans  Row mean scores differ 8.5025  1 0.0035465
cmeans  Col mean scores differ 8.5025  1 0.0035465
cor        Nonzero correlation 8.5025  1 0.0035465


$ALL
Cochran-Mantel-Haenszel Statistics for Treatment by Improvement
    Overall tests, controlling for all strata

                AltHypothesis  Chisq Df       Prob
general    General association 18.411  1 1.7807e-05
rmeans  Row mean scores differ 18.411  1 1.7807e-05
cmeans  Col mean scores differ 18.411  1 1.7807e-05
cor        Nonzero correlation 18.411  1 1.7807e-05
```

```r
mantel_fleiss(dat_cc,2)
```

```
[1] 36
```

3.9.4 Answer

- The Mantel-Fleiss criterion was determined to be 36, which means the chi-square assumption holds.
- $Q_{MH} = 18.4$, $p < 0.0001$, there is strong evidence of a conditional association between the treatment and improvement.

3.10 Homogeneity of Odds Ratios

The Breslow-Day test or the Zelen's test can be used to test if the homogeneous association is satisfied.

3.10.1 Breslow-Day test

The null hypothesis associated with the Breslow-Day test is that there is homogeneous association between X and Y.

> **Note**
>
> When the null hypothesis is rejected, there is an implied three-way association between the variables. This implies that conditional associations should be investigated at each strata.

$$Q_{BD} = \sum_{h=1}^{q} \sum_{i=1}^{2} \sum_{j=1}^{2} \frac{(n_{hij} - m_{hij})^2}{m_{hij}}$$

which approximately follows a chi-square distribution with $(n-1)$ degrees of freedom.

3.10.2 Zelen's Test

Zelen's test is similar to the Breslow-Day test, but uses exact methods.

> **Note**
>
> Similar to the Fisher's Exact Test, the Zelen's test entails calculating the probability of all possible sets of tables. The resulting p-value is the sum of all table probabilities that are lower than the probability of the observed table. This test is not available in `R`.

4 Measures of Association

4.1 Common Odds Ratio

When there is homogeneous association, we can compute a **common odds ratio** to summarize the conditional associations that exist in the partial tables.

- The Mantel-Haenszel estimator $\hat{\psi}_{MH}$ can approximate the common odds ratio such that

$$\hat{\psi}_{MH} = \frac{\sum_{j=1}^{q} n_{j11} n_{j22} / n_j}{\sum_{j=1}^{q} n_{j12} n_{j21} / n_j}$$

> **Note**
>
> We can calculate confidence intervals (asymptotic and exact) for common odds ratios!

4.2 Common Odds Ratio: Confidence Intervals

There are two confidence intervals for the common odds ratios provided by SAS: The Mantel-Haenszel and Logit confidence limits.

4.2.1 Mantel-Haenszel

- based on an estimate of the variance of $\log \psi_{MH}$
- Not sensitive to small sample sizes

4.2.2 Logit

- based on weighted regression estimates
- reasonable but requires adequate sample sizes, i.e. $n_{hij} \geq 5$

4.3 Common Odds Ratios: SAS

- The common odds ratios are calculated with the `cmh` option
  - Only includes asymptotic (MH and logit) CI limits.
  - Including the `measures` options yields plots of common odds ratios and stratum-level odds ratios.
- The Breslow-Day test is also included in the `cmh` option

- The `comor` and `eqor` options in the `exact` statement yield the exact common odds ratio and the Zelen's test results, respectively.

Sample SAS code:

```
proc freq;
tables a*b*c/cmh measures;
exact comor eqor;
run;
```

4.4 Common Odds Ratios: `R`

The `mantelhaen.test()` provides an estimate of the common odds ratio for a 2x2 contingency table. This function provides the Mantel-Haenszel confidence interval provided in `SAS`.

The `breslow_day_test()` function in the `vcdExtra` package performs the Breslow-Day test.

The `zelen.test()` function in the `NSM3` package performs the Zelen's test.

4.5 Example

The table shows the data from a study on coronary artery disease. Investigators were interested in whether electrocardiogram (ECG) measurement was associated with disease status. Gender was thought to be associated with disease status, so investigators post-stratified the data into male and female groups. In addition, there was interest in examining the odds ratios.

4.5.1 Data

Table 3.7 Coronary Artery Disease Data

Sex	ECG	No Disease	Disease	Total
Female	< 0.1 ST segment depression	11	4	15
Female	≥ 0.1 ST segment depression	10	8	18
Male	< 0.1 ST segment depression	9	9	18
Male	≥ 0.1 ST segment depression	6	21	27

4.5.2 SAS Code

```
data ca;
input sex $ ECG $ disease $ count;
datalines;
female <0.1 yes 4
female <0.1 no 11
female >=0.1 yes 8
female >=0.1 no 10
male <0.1 yes 9
male <0.1 no 9
male >=0.1 yes 21
male >=0.1 no 6
;
proc freq data=ca ;
weight count;
tables sex*disease / nocol nopct chisq measures;                    ①
tables sex*ECG*disease / nocol nopct cmh measures;                  ②
run;
```

① Testing for marginal association between sex and disease status
② Using sex as a stratification variable/strata. `measures` includes common odds ratio plots.

4.5.3 R Code

```
dat  <- expand.grid(Sex = c("Female","Male"),
                    ECG=c("Less","Greater"),
                    Disease = c("No","Yes")
                   )
dat$count = c(11,9,10,6,4,9,8,21)

dat_cc <- xtabs(count ~  Disease+Sex, data=dat)
dat_cc
```

```
       Sex
Disease Female Male
    No      21   15
    Yes     12   30
```

```
CMHtest(dat_cc, correct=F)
```

```
Cochran-Mantel-Haenszel Statistics for Disease by Sex
```

```
                AltHypothesis  Chisq Df     Prob
cor        Nonzero correlation 6.9444  1 0.008408
rmeans  Row mean scores differ 6.9444  1 0.008408
cmeans  Col mean scores differ 6.9444  1 0.008408
general    General association 6.9444  1 0.008408
```

Accounting for partial tables,

```
dat_cc <- xtabs(count ~  ECG+Disease+Sex, data=dat)
dat_cc
```

```
, , Sex = Female

         Disease
ECG       No Yes
  Less    11   4
  Greater 10   8

, , Sex = Male

         Disease
ECG       No Yes
  Less     9   9
  Greater  6  21
```

```
library(vcdExtra)
breslow_day_test(dat_cc)
```

```
Breslow-Day Test on Homogeneity of Odds Ratios

Data:          dat_cc
OR variables:  ECG, Disease
Strata:        Sex
Common OR:     2.847

X-squared = 0.2155, df = 1, p-value = 0.6425
```

```r
mantelhaen.test(dat_cc, correct=F)
```

```
    Mantel-Haenszel chi-squared test without continuity correction

data:  dat_cc
Mantel-Haenszel X-squared = 4.5026, df = 1, p-value = 0.03384
alternative hypothesis: true common odds ratio is not equal to 1
95 percent confidence interval:
 1.076514 7.527901
sample estimates:
common odds ratio
         2.846734
```

```r
library(NSM3)
```

```
Warning: package 'NSM3' was built under R version 4.6.1
```

```
Loading required package: combinat
```

```
Warning: package 'combinat' was built under R version 4.6.1
```

```
Attaching package: 'combinat'
```

```
The following object is masked from 'package:utils':

    combn
```

```
Loading required package: MASS
```

```
Attaching package: 'MASS'
```

```
The following object is masked from 'package:dplyr':

    select
```

```
Loading required package: partitions
```

```
Warning: package 'partitions' was built under R version 4.6.1
```

```
Loading required package: survival
```

```
zelen.test(dat_cc)
```

```
Zelen's test:
P = 1
```

4.5.4 Answer

- Sex was determined to be marginally associated with disease status ($\chi^2 = 6.9, p = 0.0084$).
- Controlling for sex, there is strong evidence that the ECG measurements is conditionally associated with disease status ($\chi^2 = 4.5, p = 0.03$).
- The Breslow-Day test yielded a non-significant p-value (p=0.64), hence we can use the common odds ratios
- The MH odds ratio was estimated to be 2.86 (95% CI: 1.1,7.5). We can interpret this value as: The odds of having the disease is 2.86 times that for patients with a high ECG segment depression compared to those with a low ECG segment depression.

4.6 Exercise

Table 3.8 shows the data from research done by a small company in initiating exercise programs (office/home) in their two locations in a nearby suburbs. After a year, participants and non-participants underwent a cardiovascular stress test to assess their fitness, and their result was recorded as good or not good depending on age-adjusted criteria. The exercise counselor was interested in whether type of program was associated with good test results.

- Calculate the common odds ratio ψ_{MH} and the **exact** 95% confidence intervals.
- Use Zelen's test to test if the odds ratio is uniform in both locations.

Table 3.8 Cardiovascular Test Outcomes

Location	Program	Good	Not Good	Total
Downtown	Office	12	5	17
Downtown	Home	3	5	8
	Total	15	10	25
Satellite	Office	6	1	7
Satellite	Home	1	3	4
	Total	7	4	11

4.7 Exercise

Table 3.8 shows the data from research done by a small company in initiating exercise programs (office/home) in their two locations in a nearby suburbs. After a year, participants and non-participants underwent a cardiovascular stress test to assess their fitness, and their result was recorded as good or not good depending on age-adjusted criteria. The exercise counselor was interested in whether type of program was associated with good test results.

- Calculate the common odds ratio ψ_{MH} and the **exact** 95% confidence intervals.
- Use Zelen's test to test if the odds ratio is uniform in both locations

4.7.1 SAS Code

```
data exercise;
input location $ program $ outcome $ count @@;
datalines;
downtown office good 12 downtown office not 5
downtown home good 3 downtown home not 5
satellite office good 6 satellite office not 1
satellite home good 1 satellite home not 3
;

proc freq order=data;
weight count;
tables location*program*outcome /cmh(mf);                    ①
exact comor eqor;                                            ②
run;
```

① MF should be enclosed in parentheses
② `eqor`: Zelen's test, `comor`: exact confidence interval for the common odds ratio.

4.7.2 R Code

```
dat  <- expand.grid(Location = c("D","S"),
                    Program=c("Office","Home"),
                    Outcome = c("Good","Not Good")
                   )
dat$count = c(12,6,3,1,5,1,5,3)

dat_cc <- xtabs(count ~  Program+Outcome+Location, data=dat)
dat_cc
```

```
, , Location = D

        Outcome
Program  Good Not Good
  Office   12        5
  Home      3        5

, , Location = S

        Outcome
Program  Good Not Good
  Office    6        1
  Home      1        3
```

Check the Mantel-Fleiss Criterion first.

```
mantel_fleiss(dat_cc,k=2)
```

```
[1] 4.654545
```

Because MF is less than 5, we need to use Zelen's test.

```
zelen.test(dat_cc)
```

```
Zelen's test:
P = 1
```

4.7.3 Answer

- Mantel-Fleiss criterion is not satisfied (MF=4.65), so be careful with using the asymptotic chi-square test for Q_{MH}.
  - This motivates using exact methods in calculating effect sizes.
- Common odds ratio was estimated to be 5.84 (**EXACT** 95% CI: 1.05, 33.3). There is sufficient evidence at the 0.05 significance level that the program type is associated with the quality of cardiovascular test results.
  - The Zelen's test yielded $p = 1$, which means there is no evidence against homogeneity.